WORKED SOLUTION

STEP I 2012, Question 13

Probability and Statistics • Expectation via Indicators

STEP I 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q13

The key idea

First get the conditional sample space straight: once we are told which digits are excluded, the number is just a string of five digits, each chosen from $\{1,2,3,4,5\}$, and every such string is equally likely. Then, rather than finding the whole distribution of the number of different digits, write that number as a sum of indicator variables, one for each possible digit. Linearity of expectation turns the problem into one probability, which is easy to find by looking at its complement.

Solution

The conditional sample space

Every integer from 10000 to 99999 is a string of five digits $d_1d_2d_3d_4d_5$ with $d_1 \neq 0$, and each such string appears exactly once. The condition says that every digit lies in $\{1,2,3,4,5\}$. Since 0 is excluded anyway, the restriction $d_1 \neq 0$ adds nothing, so the integers satisfying the condition correspond exactly to the

$$5^5 = 3125$$

strings of five digits from $\{1,2,3,4,5\}$.

The original choice was uniform. Conditioning on an event keeps the outcomes inside it equally likely, so given the condition each of these 3125 strings has probability $\frac{1}{3125}$. Put another way, the five digits are independent, and each is uniform on $\{1,2,3,4,5\}$: each of the 3125 outcomes is a product of five factors of $\frac{1}{5}$.

Indicator variables

Let N be the number of different digits in the choice. For $j = 1,2,3,4,5$ define

$$I_j = \begin{cases} 1 & \text{if the digit } j \text{ appears at least once,} \\ 0 & \text{otherwise.} \end{cases}$$

Each digit that appears contributes exactly 1 to the sum, so

$$N = I_1 + I_2 + I_3 + I_4 + I_5.$$

Ask yourself The I_j are not independent. Can we still add their expectations?

Yes. Linearity of expectation, $\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, holds for any random variables, independent or not. So

$$\mathrm{E}(N) = \sum_{j=1}^{5} \mathrm{E}(I_j) = \sum_{j=1}^{5} \mathrm{P}(\text{digit } j \text{ appears}).$$

The digit j fails to appear exactly when each of the five independent digits is one of the other four values, which has probability $\frac{4}{5}$ for each digit. Hence

$$\mathrm{P}(\text{digit } j \text{ appears}) = 1 - \left(\tfrac{4}{5}\right)^5 = 1 - \frac{1024}{3125} = \frac{2101}{3125}.$$

This is the same for all five values of j by symmetry, so

$$\mathrm{E}(N) = 5 \cdot \frac{2101}{3125} = \frac{2101}{625} = 3.3616,$$

as required.

Or otherwise: the full distribution

If you prefer to count directly, the number of strings using exactly k different digits is $\binom{5}{k}$ (choose the digits) times the number of ways to fill five places so that all k chosen digits appear. That second count comes from inclusion–exclusion. For example, with $k = 3$ it is $3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 = 150$.

k	1	2	3	4	5	total
strings	$5 \cdot 1$	$10 \cdot 30$	$10 \cdot 150$	$5 \cdot 240$	$1 \cdot 120$	
	5	300	1500	1200	120	3125

Then

$$\mathrm{E}(N) = \frac{1 \cdot 5 + 2 \cdot 300 + 3 \cdot 1500 + 4 \cdot 1200 + 5 \cdot 120}{3125} = \frac{10505}{3125} = 3.3616.$$

This method works, but it is longer and has more places to go wrong. The indicator method needs only one probability.

Checks

- The counts in the table add up to $3125 = 5^5$, which confirms that every string has been counted exactly once.

- The answer is sensible: N lies between 1 and 5, and with five draws from five values we expect a fair number of repeats, so a value a little above 3 is plausible. As a general check, if we draw n times from m values the same argument gives $m\big(1-(1-\frac{1}{m})^n\big)$, which is 1 when $n=1$, as it should be.
- Enumerating all integers from 10000 to 99999 by computer, keeping the 3125 that use only the digits 1–5, and averaging the number of distinct digits gives exactly $\frac{2101}{625}=3.3616$.

Common slips

- Treating the sample space as all 90000 integers. The condition reduces it to 5^5 equally likely strings.
- Thinking that linearity of expectation needs independence. It doesn't, and that is exactly why indicators are so useful.
- Computing $\mathrm{P}(\text{digit } j \text{ appears})$ by adding $\frac{1}{5}$ for each of the five places, which gives 1 because it counts strings where j appears more than once several times over. Use the complement.

Beyond STEP

This is the classic "occupancy" or coupon-collector problem: n balls thrown at random into m boxes, and N is the number of boxes that end up occupied. The indicator trick, $\mathrm{E}(N)=m\big(1-(1-\frac{1}{m})^n\big)$, is one of the most useful tools in first-year probability and in the analysis of algorithms such as hashing. The exact distribution involves Stirling numbers of the second kind, which appeared in the inclusion–exclusion count above. Blitzstein and Hwang, *Introduction to Probability*, uses indicator random variables throughout and is an excellent next step.

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